

LEDGER · 2026-08-28 · BIOGEOTECHNICS / ENVIRONMENTAL REMEDIATION

Plant-Derived Urease-Driven Enzyme-Induced Carbonate Precipitation (EICP)

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Ordinary Portland Cement (OPC)
Entry application	heavy metal immobilization and geotechnical stabilization in contaminated mining tailings and industrial soils
Measured advantage	1.83x unconfined compressive strength improvement (6.41 MPa vs 3.5 MPa for 10% Portland cement mix) at 3.3x higher raw material cost (\$52/m³ vs \$15.52/m³)
Time to first revenue	12 months
Regulatory risk	Low — no material regulatory CapEx reported (\$0 estimated)

Why it failed

- Voided by: Mismatched comparison, Economics undisclosed, Negative control / sub-2x, Form-factor / handling regression

Assessment

The report explicitly concedes failure on gross margin (45.2%, well below the 70% threshold) and primary performance delta (1.83x UCS strength vs cement, failing the 2x step-change bar). Additionally, raw material OpEx (\$52/m³) is 3.3x higher than Portland cement (\$15.52/m³), and the multi-component liquid formulation represents a strict handling regression compared to standard cement bags.

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit	Method	Ledger	Free studies	Track record	Terms
-------------------	--------	--------	--------------	--------------	-------

The Absence Audit	Method	Ledger	Free studies	Track record	Terms
-------------------	--------	--------	--------------	--------------	-------

The Absence Audit	Method	Ledger	Free studies	Track record	Terms
-------------------	--------	--------	--------------	--------------	-------

The Absence Audit	Method	Ledger	Free studies	Track record	Terms
-------------------	--------	--------	--------------	--------------	-------

The Absence Audit	Method	Ledger	Free studies	Track record	Terms
-------------------	--------	--------	--------------	--------------	-------

The Absence Audit	Method	Ledger	Free studies	Track record	Terms
-------------------	--------	--------	--------------	--------------	-------

Published for research and educational purposes. Nothing here is investment, legal, regulatory or engineering advice, and no commercial outcome is promised or implied. Concepts are drawn from published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.